

Air leak through a hull breach

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Abstract

A collection of equations and examples for precise accurate depiction of air leaks and hull breaches on spacecraft.

Section 1 presents the mass flow equation which forms the basis for the entire document, and explains how I simplified it using the assumption that the gas used is dry air.

Section 2 presents an equation for air velocity at given distance from the hole, and provides reasoning based on ideal gas law for why it is valid for air at any pressure rather than only the originally assumed atmospheric pressure.

Section 3 presents a set of 2 equations which describe how the pressure in a given pressurized volume will drop over time as a result of the puncture. I also present a simple simulation of air leakage written in python, and my observations from running it, based on which these equations were constructed.

Section 4 simply collects the 3 most important equations valid for dry air.

Section 5 applies the equations from section 4 to an example scenario, where a pressurized airtight container is left floating in a vacuum and then punctured, providing a reference use of the equations. Charts of remaining air pressure over time and air velocity at given distance from the hole are also shown.

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1 Air mass flow equation

The entire rest of the document relies on the equations provided in this section.

The equation provided below had been sourced from the paper "International space station leak localization using attitude disturbance estimation" by Jong-Woo Kim, Adam Dershowitz and John Crassidis. It is numbered 17 in the paper and located on the 5th page.

$$\dot{m} = \frac{AP\sqrt{\gamma}}{\sqrt{RT}} \left(\frac{2}{1+\gamma} \right)^{\frac{1+\gamma}{2(\gamma-1)}}$$

where:

- $\dot{m}$ is the mass flow [kg/s]
- A is the area of the hole [m²]
- P is the pressure inside the vessel [Pa]
- γ is the [specific heat ratio](#) ($\gamma = 1.4$ for dry air at 20°C)
- R is the [specific gas constant](#) of the atmosphere inside the vessel [$\frac{J}{kg \cdot K}$]
($R = 287.052874 J/kgK$ for dry air)
- T is the temperature of air on the interior [K]

I would expect "dry air" to be an extremely common assumption, ie. that the pressurized interior is filled with air at typical earth composition. This would set the constants $R = 287.052874 J/kgK$ and $\gamma = 1.4$, allowing the equation to be simplified.

Assuming $\gamma = 1.4$, the whole end of the equation can be combined into one number:

$$\sqrt{1.4} \left(\frac{2}{1+1.4} \right)^{\frac{1+1.4}{2(1.4-1)}} = 0.6847314564$$

That simplifies the original equation down to the following:

$$\dot{m} = \frac{AP}{\sqrt{RT}} \cdot 0.6847314564$$

$R = 287.052874 J/kgK$ can also be set and combined into the constant at the end, resulting in the following equation:

$$\dot{m} = \frac{AP}{\sqrt{T}} \cdot 0.04041469727 \sqrt{kgK/J}$$

- $\dot{m}$ is the mass flow [kg/s]
- A is the area of the hole [m²]
- P is the pressure inside the vessel [Pa]
- T is the temperature of air on the interior [K]

It must be remembered that this equation, after such simplification, is only valid for dry air. If one is using a different atmosphere such as for example pure oxygen, then the more complicated equation shown first should be used instead, or the simplification steps should be repeated using the γ and R for that gas instead to get a simplified equation which works for that atmospheric composition.

2 Air velocity equation

The density of dry air at room temperature and atmospheric pressure is $1.2\text{kg}/\text{m}^3$, and stays quite close to that for any temperature humans may reasonably inhabit.

The volume flow can be calculated from that, but it is only instantaneous, since the pressure and density will change as the air escapes. If the internal volume is huge you could assume that the flow will not change significantly over short times.

Even further simplifying the previous equation by assuming room temperature ($21^\circ\text{C} = 294.15\text{K}$) and atmospheric pressure ($1\text{atm} = 101325\text{Pa}$), the mass flow becomes a function of only the area of the hole:

$$\dot{m} = A \frac{101325\text{Pa}}{\sqrt{294.15\text{K}}} \cdot 0.04041469727 \sqrt{\text{kgK}/\text{J}} = A \cdot 238.7654679\text{Pa} \cdot \text{s}/\text{m}$$

To find the velocity of the air at some distance from the hole, consider that the air is moving in from all around it, which assuming that the hole is in a flat surface, means half a sphere. The volume of air being sucked in is constant, but it is being sucked through a larger and larger area (the surface of a sphere surrounding the hole) so it is moving slower the larger the distance is.

Assuming air density of $1.2\text{kg}/\text{m}^3$, the following equations describe this:

$$\dot{V} = \frac{\dot{m}}{1.2\text{kg}/\text{m}^3} = A \cdot 198.9712233\text{m}/\text{s}$$
$$v = \frac{\dot{V}}{4\pi r^2 \cdot 0.5} = \frac{A}{r^2} \cdot 31.66725372\text{m}/\text{s}$$

where:

- $\dot{V}$ is volume flow [m^3/s]
- $\dot{m}$ is mass flow [kg/s]
- A is area of the hole [m^2]
- v is velocity of the air [m/s]
- r is distance from the hole [m]

According to the ideal gas law, if all other parameters stay constant, lowering the pressure of the gas by a given factor should also decrease the density of that gas by the same factor.

In the presented equation for mass flow, decreasing the pressure x times also decreases the mass flow x times. In the later presented equation for velocity based on the assumption that air has a density of $1.2\text{kg}/\text{m}^3$, the density is presented as a hardcoded constant, but in reality decreasing the pressure by x times should also decrease the density by x times.

Since to get the velocity equation, mass flow is divided by density, and lowering the pressure decreases both of them x times, the effect actually cancels out and the velocity of the air should not change. This means that the air velocity equation is valid for air at any pressure rather than only the exact case of air at atmospheric pressure.

3 Pressure loss equation

This equation is derived using a simulation, where I simply set a big room with a given mass of air and a hole of a given size, then iterate through these 3 steps:

1. Calculate the pressure in the room in accordance with the ideal gas law (ie. assuming that air density and air pressure are proportional to each other if the volume, temperature and air composition are constant)
2. Calculate the mass flow through the hole
3. Calculate the amount of mass lost during a small amount of time
4. Subtract the mass lost from the air mass of the room

It is explained in more detail on the next page, conclusion first.

The results I observed from the simulation are well approximated by these two equations:

$$P_{now} = P_{start} \cdot 2^{-\left(\frac{t}{\lambda}\right)}$$

where:

- P_{now} is pressure at the current moment
- P_{start} is pressure at the moment the hole appeared
- t is the time that had passed since the hole appeared
- λ is the halving time of the pressure in the room

$$\lambda = \frac{V}{A \cdot \sqrt{T}} \cdot 0.05954558309 \frac{s \cdot \sqrt{K}}{m}$$

where:

- λ is the halving time of the pressure in the room
- V is the volume of the pressurized interior space
- A is the area of the hole
- T is the air temperature

The equation for mass flow is set as this:

$$\dot{m} = \frac{AP}{\sqrt{T}} \cdot 0.04041469727 \sqrt{kgK/J}$$

- $\dot{m}$ is the mass flow [kg/s]
- A is the area of the hole [m²]
- P is the pressure inside the vessel [Pa]
- T is the temperature of air on the interior [K]

Here is python code which I used to make my observations:

```
import math

def mass_flow_equation(A,P,T):
    mass_flow=A*P*0.04041469727/math.sqrt(T)
    return mass_flow
    #sanity check - if given A=0.0001,P=101325,T=293.15,it should return 0.02391682359, or something close

def room_air_density(V,M):
    return M/V

def room_air_pressure(V,M):
    return 101325*(M/V)/1.2#Pascals
    #I am assuming that if air density drops by half, the pressure also drops by half
    #if this gets volume of 1 and mass of 1.2, it should return 101325

room_volume=1#cubic metres
starting_room_air_mass=room_volume*1.2#multiply volume by 1.2kg/m^3 to get the air mass at 1 bar
hole_size=0.0001#square metres, 0.0001m^2=1cm^2
step_size=0.001#seconds, by default I have it set to 0.001 which is 1 millisecond
temperature=293.15#Kelvins, 293.15K=20 Celsius
starting_air_pressure=room_air_pressure(starting_room_air_mass,room_volume)
#given these start

#start simulation

pressure=starting_air_pressure
mass=starting_room_air_mass
mass_flow=0
mass_lost=0
time_passed=0
density=room_air_density(room_volume,mass)

while density/1.2>0.5:
    mass_flow=mass_flow_equation(hole_size,pressure,temperature)
    mass_lost=mass_flow*step_size#mass lost during one physics step
    mass-=mass_lost
    time_passed+=step_size
    density=room_air_density(room_volume,mass)
    pressure=room_air_pressure(room_volume,mass)
    if math.fmod(time_passed,1)<=step_size:
        print(pressure,time_passed)

print(f"lost half of air pressure in {time_passed} seconds")
```

Observations:

- A room of volume= $1m^3$ at 1 atmosphere (101325 pascals) with a hole of area= $0.0001m^2$ takes ≈ 34.778 seconds to drop down to 50% of the starting pressure
- That same room takes ≈ 69.555 seconds to drop down to 25% of the starting pressure, which is twice as long - this seems to confirm my assumption that the pressure loss will be well approximated by a halving time
- If the volume of the room is increased by a factor of 10, time to lose half pressure also grows by a factor of 10.
- If the area of the hole is increased by a factor of 10, time to lose pressure drops by a factor of 10.
- If the temperature is increased by a factor of 4, the time to lose pressure grows by a factor of 2. Something feels wrong about this, but it agrees with the equation.

4 Final equations for dry air

Here I'm just collecting the equations for convenience.

The velocity of air near a hole in a flat wall directly to vacuum:

$$v = \frac{A}{r^2} \cdot 31.66725372 m/s$$

where:

- $\dot{V}$ is volume flow [m^3/s]
- $\dot{m}$ is mass flow [kg/s]
- A is area of the hole [m^2]
- v is velocity of the air [m/s]
- r is distance from the hole [m]

Halving time of pressure in a punctured pressurized space:

$$\lambda = \frac{V}{A \cdot \sqrt{T}} \cdot 0.05954558309 \frac{s \cdot \sqrt{K}}{m}$$

where:

- λ is the halving time of the pressure in the room
- V is the volume of the pressurized interior space
- A is the area of the hole
- T is the air temperature

Interior pressure at given time since the punctured appeared:

$$P_{now} = P_{start} \cdot 2^{-\left(\frac{t}{\lambda}\right)}$$

where:

- P_{now} is pressure at the current moment
- P_{start} is pressure at the moment the hole appeared
- t is the time that had passed since the hole appeared
- λ is the halving time of the pressure in the room

5 Example scenario

A completely airtight container is left floating in a vacuum, which has one cubic metre of volume on the inside. It is full of dry air at atmospheric pressure and room temperature. It then gets punctured, with the area of the hole being one square centimetre.

Pressure halving time:

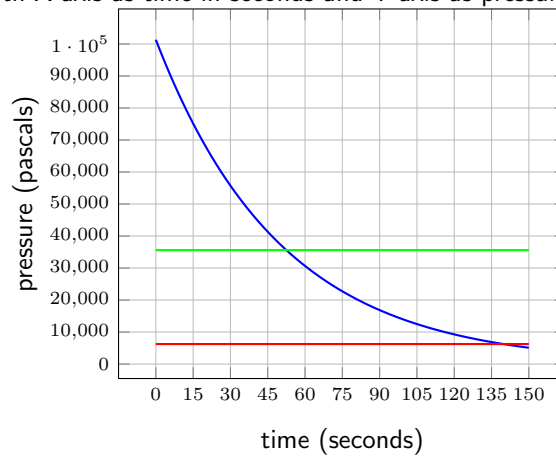
$$\lambda = \frac{1m^3}{0.0001m^2 \cdot \sqrt{293.15K}} \cdot 0.05954558309 \frac{s \cdot \sqrt{K}}{m} = 34.778s$$

Pressure will drop by half after 34.778 seconds, after twice that time it will be one quarter, after thrice that time it will be one eighth, etc.

Example, what will the pressure be at exactly 15 seconds since the container was punctured:

$$P_{now} = 101325Pa \cdot 2^{-\left(\frac{15s}{34.778s}\right)} = 75141.56Pa$$

Graphed with X axis as time in seconds and Y axis as pressure in pascals:



The climber's "death zone" (minimum air pressure needed to maintain consciousness without supporting equipment) is marked as a green horizontal line, and the armstrong limit (ie, pressure low enough that the boiling points of bodily fluids drops below typical human body temperature, necessitating full pressure suits) is marked as a horizontal red line.

Velocity of air getting sucked into the hole at 5cm distance while at full atmospheric pressure:

$$v = \frac{0.0001m^2}{(5cm)^2} \cdot 31.66725372m/s = 1.266690149m/s$$

Graphed with X axis as distance from the hole in centimetres and Y axis as velocity in m/s:

